Codecs: Separating Meaning from Language

Coherent dialogue in thought-vector space from 48M parameters on one consumer GPU

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July 19, 2026

Abstract

We present a ∼48M-parameter conversational system, trained from scratch in ∼25 GPU-hours on a single consumer GPU (AMD RX 6700 XT), that conducts dialogue entirely in a learned sentence-latent space: no token-level language modeling occurs in its reasoning component. A 32.9M-parameter *codec* encodes text as a sequence of "thought vectors" ordered by importance, so any prefix decodes back to text and compression ratio becomes a decode-time choice (byte-perfect at 4:1, readable at 8:1). A 15.1M-parameter *thinker* predicts the reply's thought vectors from the conversation's; the codec decoder renders the words. The result is coherent, grounded, context-sensitive small talk from a system orders of magnitude smaller than typical dialogue LLMs — evidence that conversational competence and token generation are separable concerns — and because the thinker never touches tokens, codecs could be added per language, domain, or modality while the thinker is reused. We also chart what this scale does not achieve: a persistent *sentiment-register* failure (cheerful replies to bad news), traced via latent-space diagnostics to a data absence — no training conversation ever reverses mood mid-dialogue. In-distribution data patching defeated the register *probes* (contextual error 0.50 → 0.17) but bought only partial, topic-gated routing in live chat. A pre-registered comparison against a parameter-, data-, and compute-matched token LM splits cleanly: the token LM wins per-turn conversational quality; the latent system wins reply cost and context memory, which stay flat in conversation depth where the LM's grow with history; and the matched LM reproduces the same contextual register failure — evidence the root cause is the data, not the architecture. Four documented incidents of models satisfying lexical metrics without the behavior they proxy round out the record.

1 Introduction

Large language models converse by autoregressing over tokens, entangling understanding, reasoning, and rendering in a single token-level model. An alternative line of work asks whether the *semantic* content of an utterance can be manipulated as a small set of continuous vectors, with token generation delegated to a decoder [9, 4, 2]. We[1] push this question to an extreme: a ∼48M-parameter system, trained from scratch on one consumer GPU, in which dialogue happens in latent space end-to-end — and it works. The system holds coherent, grounded, context-sensitive small talk with no token-level language modeling anywhere in its reasoning component. The design instinct throughout is mechanical: the fewest moving parts that still do the job — one codec, one thinker, and a frozen interface between them. We take this as evidence that conversational competence and token

[1] "We" in this paper is one human and a rotating cast of AI instances; who did what is spelled out in the contribution statement before the acknowledgments.

generation are separable concerns, with direct consequences for how such systems could scale and specialize (§7).

Contributions:

1. **A working latent-space conversationalist at 48M parameters.** A thinker transformer that maps context thoughts to response thoughts under a winner-take-all multi-hypothesis loss with random-prefix targets and cycle consistency, with the codec decoder anchored against drift, producing coherent grounded multi-turn dialogue (§4, §6).

2. **Prefix-decodable ordered latents.** A text autoencoder whose latent sequence is ordered by importance so that every prefix is a valid, gracefully degrading representation, plus a predictor that turns this into an adaptive-compression dial (§3).

3. **A sharp negative on conditioning.** A case study tracing a behavioral disease to a data-distribution absence via latent-space diagnostics — through training levers, fine-tuning, and in-distribution data patching, none of which produce more than partial, topic-gated routing (§6.2).

4. **The full experimental record.** Eight transcript-audited ablation rounds, including negative results, and four incidents of metric gaming (§6.4).

2 The system in brief

An LLM does three jobs with one network — understand the input, decide what to say, and put it into words — all by predicting the next token. This system splits those jobs apart (Figure 1). The **codec** is a text autoencoder: its encoder compresses a sentence into a short sequence of vectors — *thought vectors* — and its decoder turns thought vectors back into text. The **thinker** is a small transformer that reads the thought vectors of the conversation so far and predicts the thought vectors of the reply; the codec decoder then renders those into words. The thinker never sees a token: understanding and response choice happen entirely in vector space, and the decoder only does the wording.

The codec's vectors are ordered by importance: the first few carry the gist, later ones add detail, and *any* prefix decodes to sensible text. That makes compression a decode-time dial — reconstruction is byte-perfect keeping one vector per four tokens, still readable at one per eight — and it lets the thinker work from just the first eight thoughts of each turn, where a token LM would attend over every token of every turn.

The whole system is ∼48M parameters (32.9M codec, 15.1M thinker), trained from scratch in ∼25 GPU-hours on one consumer GPU (AMD RX 6700 XT). What that buys, verbatim from the flagship at temperature 0:

```
user > i'm feeling really overwhelmed with work lately.
bot  > I can imagine. What's been going on?
user > my boss keeps piling on deadlines.
bot  > That sounds like a lot of pressure. Have you been
       able to talk to anyone else?
```

It holds grounded, context-sensitive small talk — follow-up questions, multi-turn reference — within a small-talk domain; the same conversation slips later in the run (Appendix B shows it unedited). Its one well-mapped failure: after upbeat turns it tends to answer bad news cheerfully, a data-distribution problem dissected in §6.2. Headline numbers are in Tables 1 and 2.

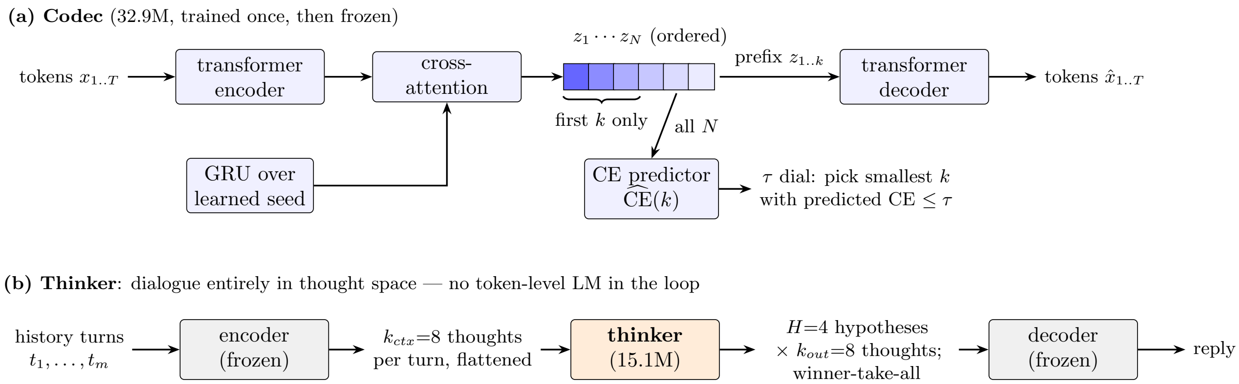

Figure 1: System architecture. **(a)** The codec encodes text into N importance-ordered thought vectors: a GRU unrolled over a learned seed produces ordered query slots that cross-attend the transformer-encoded tokens. Any prefix $z_{1..k}$ decodes back to text; a predictor head estimates reconstruction CE per prefix length, so compression ratio is a decode-time dial (τ). **(b)** The thinker converses in that latent space: each history turn becomes k_{ctx} thoughts, the thinker predicts k_{out} response thoughts in H winner-take-all hypotheses, and the frozen decoder renders the selected hypothesis as text.

2.1 Running it

Chat in the browser at the hosted demo,[2] or locally — CPU is enough:

```
git clone https://github.com/nochinator/thought-vectors
cd thought-vectors && scripts/setup_env.sh --cpu
# from the GitHub release (the thinker embeds codec weights but
# reads codec config from the codec file - you need both):
#   FINAL_12H-best.pt   -> checkpoints/FINAL_12H/best.pt
#   m5_frontier-best.pt -> checkpoints/m5_frontier/best.pt
.venv/bin/tv-chat --ckpt checkpoints/FINAL_12H/best.pt --device cpu
```

The rest of the paper is the detail: how the codec and thinker are built and trained (§3–4), what the experiments found and where the limits are (§6), and what to try next (§7.1).

3 The codec: ordered, prefix-decodable thought vectors

The codec is a text autoencoder: a transformer encoder processes the token sequence; a GRU unrolled over a learned seed produces N *ordered* query slots that cross-attend the encoded tokens, yielding thought vectors $z_{1..N}$ (d=384); a transformer decoder generates tokens conditioned on a *prefix* $z_{1..k}$ (Figure 1). Training samples k randomly per example (with a compression-weighted schedule), which forces the slots into an importance ordering: early slots carry the gist, later slots refinements. A small predictor head estimates reconstruction CE for each prefix length, enabling decode-time adaptive compression (τ dial). Total 32.9M parameters (19.0M encoder, 20.1M decoder, 6.2M shared token embeddings).

Headline behavior (M5 frontier run, 55,140 steps / 12h; eval on a 2K-text held-out mix, batched greedy decode):

[2] https://huggingface.co/spaces/nochinator/thought-vectors-chat

Setting	Ratio	Reconstruction	Notes
r=0.25	4:1	F1 1.00, CE ∼0.001	byte-perfect, every length bucket (≤257 tok)
r=0.125	8:1	F1 0.83–0.96	graceful: adjective swaps, tail truncation
τ=0.25	∼6:1	CE 0.02–0.08	adaptive: predictor picks k per text
τ=2.0	∼12:1	CE ∼1.7	gist-level, still readable
k=64 fixed	—	perfect ≤257 tok	predictor MAE 0.05 at k=32

Table 1: Codec reconstruction across the compression dial. The quality/ratio trade-off is monotone and graded; there's no garbage regime.

Against the earlier prototype. The project's first codec (a d=256 GRU autoencoder; its full log is preserved at `legacy/phase0/RESEARCH.md`) plateaued at 50–62% word overlap beyond ∼80 tokens *no matter how many latent vectors the decoder received.* The present codec removes that ceiling (byte-perfect at 4:1 through 257 tokens, Table 1) with only d=384, through training-side changes alone — the prototype's limits were recipe artifacts, not limits of the thought-vector idea.

4 The thinker: conversing in thought space

Dialogue history turns are each encoded to k_{ctx}=8 thoughts. The thinker (15.1M params) is a pre-norm transformer over the flattened context thought sequence with role, turn-distance, and slot-position embeddings. A GRU-scaffolded query head cross-attends the trunk to emit k_{out}=8 response thoughts in parallel, in H=4 winner-take-all hypotheses: only the hypothesis closest to the encoded gold response receives gradient [3]. Random-prefix targets (k sampled per batch, $k_{\min}$=4) align training with the codec's prefix decodability.

Two additions proved decisive in ablation: **(i) cycle consistency** — decode the predicted thoughts, re-encode the text, and penalize distance to the prediction (applied to 25% of batches, w=0.5); **(ii) decoder co-training** — unfreeze the codec decoder at 0.05× LR with a compression-anchor loss on 40% of batches so it adapts to imperfect predicted thoughts without forgetting the codec geometry. At inference the reply is decoded from the selected hypothesis (most-decodable by default; §6.2 introduces a last-turn affinity rule).

5 Experimental setup

Hardware. One AMD RX 6700 XT (12 GB, gfx1031/ROCm), consumer desktop. All chat and CPU evals ran on the host CPU (AMD Ryzen 5 5600G, 32 GB RAM); HIP inference is broken on this GPU architecture (gfx1031).

Data. Dialogue mix: SODA (61K conversations), OASST1 best-ranked English paths (8.5K), PersonaChat (0.9K); later rounds add EmpatheticDialogues (23K) and 40K synthesized *register-reversal* splices (§6.2). Turn-aware shards; role by turn parity; every non-first turn is a training target.

Metrics. Latent cosine to the gold response (val_cos), decoded CE, reference F1, distinct-n; behavioral probes: *self-repetition* (pairwise unigram-F1 among a model's own replies), *context sensitivity* (F1 between with-history and without-history replies), and a *sentiment-register* probe suite (bad news, good news, and contextual bad-news-after-good-news; positive-affect lexicon match).

All lexical metrics are treated as *defeasible*: no metric win is accepted without a transcript audit and a live chat probe (§6.4).

Ablation protocol. Rounds of 45-minute arms at fixed wall time, warm-started from the current flagship for fine-tuning rounds or trained from scratch for recipe rounds; pre-registered decision rules per round.

Total compute. The ∼25 GPU-hours in the abstract are what the released weights cost, not what the findings cost. Behind them sit codec bracket studies, eight thinker ablation rounds, frontier predecessors, and diagnostics — some 80 short arms (15–45 minutes each) and four half-day runs across 86 checkpointed runs in this phase, on the order of another hundred GPU-hours on the same device, unmetered precisely. The two earlier prototype generations (see the project history) add more still. The matched-baseline round (§6.5) adds ∼36 GPU-hours: thirteen 45-minute recipe arms and one 24-hour run.

6 Results

6.1 Flagship runs

Run (12h thinker)	val_cos↑	ref_F1↑	self_rep↓	ctx_sens↓	reg↓	reg_ctx↓	pos_ok↑
FINAL_12H (original mix)	**0.428**	**0.297**	0.188	**0.146**	0.17	0.50	0.17
FINAL2_12H (+ED +splices)	0.415	0.278	**0.154**	0.206	0.00*	**0.17***	**0.50**

Table 2: Flagship comparison; both thinkers trained 12h from scratch on the frozen M5 codec — the only difference is the data mix. Register scores are lexicon-scored; *hand audit of the FINAL2_12H transcripts finds errors the lexicon misses (honest reg ≈ 0.25, reg_ctx ≈ 0.33; §6.2). Round-trip guard: PASS for both.

FINAL_12H (thinker trained 12h from scratch on the frozen M5 codec, cycle + decoder co-training, original mix) achieved the best values of every metric across ∼30 arms: val_cos 0.428, dec_CE 2.69, ref_F1 0.297, self_rep 0.188, ctx_sens 0.146, round-trip guard PASS. Qualitatively: grounded openings, follow-up questions, multi-turn reference. FINAL2_12H (identical recipe, data mix + EmpatheticDialogues + reversal splices) trades a small amount of general quality (val_cos, ref_F1, ctx_sens) for large gains on the register probes; §6.2 shows only part of those gains survives live-chat audits — partial, topic-gated routing — so FINAL_12H remains the flagship on overall quality.

6.2 Case study: the sentiment-register disease

The flagship's failure was conditional, not constant: bad news on its own was usually answered correctly (register error 0.17), but after an upbeat turn, bad news drew *cheerful* replies — "That's great!" — tripling the error rate (contextual error 0.50).

Round 6 (training levers): negative. Every arm — including a do-nothing control — *worsened* contextual register (0.50 → 0.67–0.83). Continued exposure to relentlessly upbeat small talk is itself the disease; no loss or architecture lever fixed a data problem.

Round 7 (empathetic data): style without routing. Mixing EmpatheticDialogues transferred the commiseration *style* ("Oh no! I'm so sorry."; positive replies to good news rose

$0.17 \rightarrow 0.83$) but register *routing* worsened. Three diagnostics located the failure: (i) the frozen encoder separates sentiment (83% nearest-centroid accuracy), but positive thoughts form a tight cluster while negative thoughts are diffuse — positivity is a compact attractor; (ii) selecting the hypothesis whose pooled thought best matches the *last user turn* (a zero-training inference change) halves single-turn errors; (iii) decoding *all* hypotheses after an upbeat turn shows every hypothesis is positive — winner-take-all diversity is style-level, not register-level, so no selection rule can fix contextual register.

Root cause. Every training conversation — in all corpora used — holds a single mood throughout. Mid-conversation register reversal has zero support in the training distribution; the trunk learned conversation-level mood conditioning, which is exactly the observed disease.

Round 8 (reversal splices). We synthesize reversal dialogues by splicing EmpatheticDialogues conversations by emotion label (positive opening exchange + negative continuation; 25% flipped direction). Fine-tuning produced the first contextual commiserations observed in any arm, but degraded single-turn routing — fine-tuning transfers style, not conditioning. FINAL2_12H therefore trains from scratch with reversals in-distribution ($\sim$30% of turns from splices).

FINAL2_12H: the probes fall; the skill comes back partial. On the register suite FINAL2_12H posts the best scores of any checkpoint (lexicon: 0.00 single-turn, 0.17 contextual; hand audit: $\approx$0.25 / $\approx$0.33, still the best contextual score observed), including the first correct contextual reversals at temperature 0 ("oh no. what happened?" after an upbeat wedding-planning turn). But the mandatory live-chat probe, using a reversal conversation of the same *shape* as the splices with novel phrasing (beach vacation $\rightarrow$ apartment burglary), reproduced the disease exactly: "I bet that was fun!" to the break-in. That verdict rested on a four-conversation probe, so we widened the audit to a ten-reversal battery of novel topics, pivot positions, and pivot cues (Appendix C). The battery locates the truth between the metric and the burglary transcript: FINAL_12H commiserates on *zero* of the ten, FINAL2_12H on three (four to five under affinity selection, §6.3, which also corrects the burglary probe itself), and FINAL2's hits cluster on EmpatheticDialogues' dense distress topics — illness, job loss, a lost pet — while situational and property misfortune (a burglary, a hailstorm, a rear-ending) still draw cheer. The patch bought *partial, topic-gated* routing: the model reads strong distress vocabulary, not last-turn sentiment. Together with Rounds 6–8 the case study still closes negative, with a sharper boundary: at this scale and objective, neither training levers, nor fine-tuned style transfer, nor in-distribution data patching produces *reliable* register routing — the patch generalizes only where its distress vocabulary is dense. One untested candidate for the residual gap is architectural rather than data-side: the thinker consumes history as a single flattened sequence of per-turn thought prefixes, and a trunk that treated within-turn slot order and across-turn time as distinct axes might support conditioning the flat view does not — larger than this project's memory budget (§7.1).

The disease is paradigm-independent. The matched token LM of §6.5, trained 24 hours on the same mix, commiserates on one of the battery's ten reversals (hand-audited contextual error 0.50 — FINAL_12H's exact level) and at its endpoint still answers "the test results came back and it isn't good news" with "That's fantastic news!". Sustained-mood register, by contrast, was already correct two hours into its training. The failure that transfers across paradigms is precisely contextual routing — the pre-registered prediction this round existed to falsify — which moves the root-cause diagnosis from consistent with the data to tested against a matched control from the other paradigm.

6.3 Zero-training win: affinity hypothesis selection

Selecting the WTA hypothesis by pooled-thought cosine to the last user turn (instead of decodability) cut the flagship's single-turn register errors from 0.17 to 0.08 with no retraining, at no cost elsewhere; on FINAL2_12H it lifts pos_ok 0.50 → 0.67 at equal contextual error, and on the reversal battery of Appendix C it adds one to two contextual commiserations and converts most remaining cheerful misses to neutral.

Checkpoint	Hypothesis selection	reg↓	reg_ctx↓	pos_ok↑
FINAL_12H	decodability	0.17	0.50	0.17
FINAL_12H	affinity	**0.08**	0.50	0.17
FINAL2_12H	decodability	0.00	0.17	0.50
FINAL2_12H	affinity	0.08	0.17	**0.67**

Table 3: Register metrics under the two WTA selection rules (lexicon-scored, temperature 0). Affinity selection is a pure inference-time change.

6.4 Metric gaming: four incidents

Four times, a model satisfied a metric without the behavior it proxies: (1) an arm minimized context-sensitivity by adding *noise* rather than using context; (2) an arm "won" the contextual register probe by phrasing cheer in words missing from the positive-affect lexicon ("terrific", "beautiful"); (3) a later arm repeated the trick with "so sweet"; (4) FINAL2_12H scored a perfect 0.00 single-turn register while its transcripts contained "How are you going to celebrate?" in reply to a sprained ankle. Each was caught by the audit rule of §5. We widened the lexicon twice, re-scoring all checkpoints each time; after the fourth incident we stopped widening and instead report hand-audit corrections alongside lexicon scores (Table 2).

6.5 Against a matched monolithic baseline

The obvious question is whether a plain token LM at the same cost does better. We pre-registered the comparison (predictions, decision rules, and a fixed tuning budget written down before any run; `docs/BASELINE_ABLATIONS.md`) and gave the baseline a larger recipe search than the thinker ever had: thirteen 45-minute recipe arms searched shape, learning rate, batch, and context window, and the winner — a textbook decoder-only transformer, d 640 × 8 layers, tied embeddings, 50.1M parameters, the same tokenizer, the same data view (loss on every non-first turn), the same GPU — then trained for 24 hours, the codec's 12 plus the thinker's 12. Both systems were scored by the identical eval scripts and probes.

Two of the three pre-registered predictions held; the quality prediction did not. The token LM wins per-turn conversational quality outright — better reference F1, twice the lexical diversity, less self-repetition, stronger context sensitivity, and transcripts that back the metrics — where we had predicted the thinker would hold grounded coherence at this scale. The efficiency prediction held: reply cost is tied at one turn and the latent system is ∼2.5× cheaper and flat from five turns, with ∼190× smaller context state. The register prediction held in its contextual form (§6.2): the LM's lexicon scores look clean (0.00 / 0.17), but hand audit finds contextual error 0.50, one battery commiseration, and "That's so cool!" to a stolen wallet — the metric-gaming pattern of §6.4 recurs on the baseline side. The LM also shows a failure the thinker never did: an *inverse* register error, answering three of six good-news probes with "i'm sorry to hear that" — its default register is

	FINAL_12H (48.0M)	token LM (50.1M)
ref_F1↑	0.297	**0.342**
distinct-1 / 2↑	0.032 / 0.133	**0.067 / 0.286**
self_rep↓	0.188	**0.170**
ctx_sens↓	0.146	**0.114**
reg_ctx↓ (lexicon / hand)	0.50 / —	0.17 / 0.50
battery commiserations	0/10	1/10
reply cost, 1 turn (GFLOP)	1.35	1.45
reply cost, 10 turns	**3.0**	7.6
reply cost, 20 turns	**3.0**	7.4
context memory, 20 turns (floats)	**18.4k**	3.69M

Table 4: Matched comparison at ∼50M parameters, same data, tokenizer, and total training compute. Quality metrics as in Table 2; reply cost is measured KV-cache-equivalent FLOPs (one teacher-forced pass over context + reply, the same convention both sides); context memory is what each system carries between turns. The LM's context window caps at 384 tokens, so its cost plateaus at ∼8 short turns by *forgetting*; the thinker's plateaus at 6 turns by design, carrying $8 \times d$ floats per turn.

apology where the thinker's was cheer, and neither routes on last-turn sentiment. At temperature 0 it also produced one verbatim repetition loop on the battery, and its transcripts leak assistant boilerplate ("As an AI language model...") from the synthetic side of the shared data.

Free-form conversation, driven live rather than scripted (transcripts in `logs/livechat_roundb_20260715.txt`), shows an asymmetry the metrics do not capture: the two systems fail in different *directions*. The LM's errors compound — its own generated tokens return as context, so one odd reply raises the odds of the next, and twice in ten conversations it entered a verbatim repetition loop it never left. The latent system structurally cannot follow that path: the thinker never sees its own tokens, only the codec's compressed re-encoding of them, and surface repetition does not survive the compression well enough to self-reinforce — its errors are independent per turn (abrupt topic jumps) rather than a random walk with drift. Sampling noise mirrors the split. Temperature perturbs the LM's conversational state directly, token by token; the thinker's thought vectors are deterministic, so temperature can attack only the rendering — but it attacks through a decoder conditioned on lossy vectors (val_cos 0.428), the system's one fragile layer, which degrades into non-words where the LM stays grammatical-but-wrong. The latent system pairs a stable conversational state with a brittle surface; the LM pairs a higher per-turn ceiling with an unstable state. The parameter split explains the ceiling difference: the LM's full 50.1M parameters sit in the reply path, while the latent system selects its reply with the thinker's 15.1M and spends the other 32.9M on rendering — the asymmetry §7 builds on. Both systems share the domain's ceilings: neither can answer "what did I say I was doing yesterday?", and neither recovers from a correction — history tints mood and topic; it is not retrievable knowledge.

At matched cost the monolithic LM is the better conversationalist per turn. The latent system's case rests on the measured curves — reply cost and context memory that do not grow with conversation depth — on the codec-swap properties of §7, and on the register result, which localizes the paper's central failure in the data rather than in either architecture.

7 Outlook and further work

The headline finding — coherent dialogue from 48M parameters with no token-level language modeling in the reasoning loop — matters mostly for what it implies at larger scale. This section lays out those implications; they are directions, not demonstrated results.

Parameter and sequence economy. The codec and the thinker scale on different axes: a codec must cover the surface forms of its input distribution — a need that grows slowly once the language is covered — while thinker capacity governs the quality of thought prediction itself. In a scaled system, added parameters go almost entirely to the thinker rather than being spent in every layer re-modeling token statistics; our results are consistent with the codec saturating early (byte-perfect at 4:1 throughout, while the 15.1M thinker binds quality). The live-chat asymmetry of §6.5 is this argument in miniature: the matched system's deficit is thinker knowledge, while its stability comes from the latent bottleneck — added parameters belong in the thinker. Sequence cost shifts the same way, twice over. First, the representation itself is denser: a token LM spends one embedding per token, while the codec carries the same text byte-perfectly at one vector per four tokens (Table 1) — and since self-attention cost is quadratic in sequence length, a 4× shorter sequence is up to ∼16× cheaper to attend over at equal fidelity. Second, this system compresses further still: the thinker sees a fixed k_{ctx}=8 vectors per turn whatever the turn's length and emits a reply in one forward pass, so context cost grows with *turns*, not tokens, and generation latency moves to a small decoder. The codec is the overhead that buys this: each turn is encoded once and each reply decoded once — a fixed per-turn cost that does not grow with history. At our scale that overhead is most of the system's parameters, so the net saving today is modest; in a scaled system the codec cost stays fixed while the attention savings grow with both history length and thinker size.

Swappable codecs. The thinker never sees tokens, and nothing in its architecture is specific to text. Codecs could be trained per modality (speech, images, structured data) or per domain (languages, registers, programming languages), each small and cheap, all feeding one shared thinker: multimodality and specialization become codec-training problems rather than retrain-the-reasoner problems — though aligning multiple codecs into one shared thought geometry is itself an open problem.

7.1 A roadmap by compute budget

Almost every boundary this paper reports is a compute boundary, not a conceptual one, so this roadmap is organized by what an experiment costs. Every item traces to a documented observation (cited in place), and every item has a concrete starting point in the repository. Nothing in the first three tiers needs more than the single consumer GPU this project ran on.

No GPU: evaluation and inference.

- *A real register benchmark.* The current suite is 24 lexicon-scored items that share surface structure with the training splices (§8); the ten-reversal battery (Appendix C) is a first diversification, scored by a single reader. A larger, structurally diverse, multi-rater benchmark would make every register claim in this paper sharper. Start from `scripts/register_battery.py` and the battery transcripts in `logs/`.
- *Hypothesis-selection rules.* Affinity selection was a free win (0.17 → 0.08 single-turn register error, §6.3) and its known ceiling is documented — after an upbeat turn, all four hypotheses

decode positive, so no selection rule alone fixes contextual register — but the space of selection rules, and their interaction with temperature, is otherwise unexplored. Start from `src/thoughtvec/chat.py`.

Hours: fine-tunes on the released checkpoints.

- *Contrastive-sentiment codec unfreezing.* Diagnostic (i) of §6.2: positive thoughts form a compact attractor while negative thoughts are diffuse. Unfreezing the codec with a contrastive sentiment objective would give negative thoughts structure the thinker can route on — conditioning needs representation help, not just data.

- *The deferred VAE.* The codec trains with `kl_beta`=0; the experimental log's plan — a VAE objective as a fine-tune on the working autoencoder — was deferred for budget and never run. A smoother, sampleable latent space is untested territory.

- *Genuine reversal data.* The register patch used synthetic splices; its gains were gated to the patch data's dense distress vocabulary (Appendix C). A corpus of *naturally occurring* mid-conversation mood reversals would test whether topic-gating is an artifact of the splices or a deeper limit.

- *Longer fine-tune arms.* Every ablation arm was 45 minutes, which §6.2 shows is enough to move style but not conditioning — so this paper's fine-tuning negatives are budget-qualified. Multi-hour arms would either overturn or harden them.

Half-days: from-scratch runs.

- *A wider thinker on a longer budget.* A matched-step comparison found thinker width a tiny metric lever at the 12h budget ($\sim$+0.005 val_cos, slight coherence regression) but flagged the larger trunk as "may finish maturing" on a longer run — the experimental log explicitly designates $\sim$30M the direction for a $\gg$12h frontier. This is the single most direct scale test available.

- *A longer-sequence codec.* Byte-perfect reconstruction holds through 257 tokens at d=384 (Table 1); the cap was hardware, not architecture. Paragraph- and document-scale codecs are the obvious next bracket. Start from `configs/m5_frontier.yaml`.

- *Revisable prediction.* Single-shot prediction is the bottleneck for one-to-many dialogue: iterative-refinement (diffusion-style) heads or autoregression over thought slots would let the thinker revise a hypothesis against context instead of committing in one step — aimed directly at the all-hypotheses-positive failure of §6.2.

Beyond one GPU. The scaling implications of §7 are all open: aligning codecs for other modalities (speech, images, structured data) into one thought geometry; per-language and per-domain specialist codecs feeding a shared thinker; and scaling the thinker trunk itself, which our results suggest is the binding constraint while the codec saturates early. A fourth, motivated by §6.2: a factorized thinker that treats within-turn slot order and across-turn time as separate input axes — a two-dimensional context of turns $\times$ slots, correlated over each axis by a different mechanism, in place of the current flattened-prefix view. This is a candidate for the conditioning structure the flat trunk lacks, but training it exceeds a single consumer GPU's memory.

Reproduction instructions for the baseline system are in `docs/REPRODUCE.md`; the experimental log (`RESEARCH_LOG.md`) records the observation behind every item above, so a newcomer can check our reasoning before spending their compute on it.

8 Limitations

Scale: 48M parameters, small-talk domain; no knowledge tasks. Evaluation: register probes are few (24 items) and lexicon-scored, and the "hand" audits were transcript readings by the project's AI instances (see the contribution statement); the human author verified the final flagships' behavior in live chat but did not independently re-score the intermediate audits. The codec is frozen during thinker training (by design), capping how much the thought geometry can adapt. Single-GPU wall-time budgets bound every ablation to 45 minutes, which §6.2 shows is enough for style but not conditioning — fine-tune conclusions are qualified accordingly. The register evaluation suite shares surface structure with the reversal training splices, which is what allowed FINAL2_12H to defeat it while holding only partial routing; the ten-reversal battery (Appendix C) is a first structurally diverse check, and a larger held-out register benchmark remains future work. Finally, the chat probes are scripted and small; conclusions about live behavior rest on dozens, not thousands, of conversations, and the battery itself was scored by a single AI reader. The matched-baseline comparison (§6.5) is one 24-hour run at one scale with no seed replicates; its recipe search, though pre-registered and larger than the thinker's own, spent ~9 GPU-hours, and reply-cost curves were measured on reference implementations of both systems rather than optimized inference stacks (the KV-equivalent convention corrects the largest distortion but not all of them).

9 Related work

Sentence-level latents. Decoding text from a single continuous sentence vector goes back to sentence VAEs [1]. SONAR learns fixed-size multilingual/multimodal sentence embeddings that support generation [2], and Meta's Large Concept Models generate at the concept (sentence) level over SONAR embeddings [9] — in spirit, the closest existing system to this one. THOUGHTVECTORS differs in using a *variable-length, importance-ordered* latent sequence with prefix decodability rather than one fixed-size vector per sentence, and in targeting dialogue at roughly 1000× smaller scale due to hardware constraints.

Ordered, truncatable representations. Nested dropout trains autoencoders whose latent dimensions form an importance hierarchy, so truncation degrades gracefully [12]; Matryoshka representation learning does the same for modern embedding vectors [8]. Both order the *dimensions of one vector*; THOUGHTVECTORS orders a *sequence of vectors* (any prefix $z_{1..k}$ decodes to text), pairs the ordering with a per-prefix quality predictor so the truncation point can be chosen adaptively at decode time, and induces the ordering with per-sample random-k training rather than a dropout schedule. The encoder's learned-query cross-attention resembles Perceiver-style latent bottlenecks [5], with a GRU scaffold over the queries supplying the sequential inductive bias that makes the ordering emerge.

Latent-space reasoning. Chain-of-continuous-thought approaches feed hidden states back as inputs, keeping reasoning inside the language model's own representation stream [4]. Our thinker instead operates in an *external, frozen* codec space: the reasoning model and the text interface are separate artifacts, which is what makes the codec swappable (§7).

Multiple-hypothesis learning. Winner-take-all / multiple choice learning trains several heads and backpropagates only through the best one, capturing one-to-many mappings [3, 10]. We apply it to response-thought prediction and analyze its failure mode at this scale: hypothesis diversity turns out to be style-level, not register-level (§6.2).

Cycle consistency. Popularized for unpaired image translation [14]; we use decode–re-encode consistency as a regularizer that keeps predicted thoughts on the codec's decodable manifold.

Dialogue corpora and empathy. SODA [6], PersonaChat [13], OASST1 [7], EmpatheticDialogues [11].

10 Conclusion

A 48M-parameter system trained on one consumer GPU can hold coherent, grounded, context-sensitive small talk entirely in a learned thought-vector space, with token generation confined to a frozen decoder. Conversational competence, at least at the small-talk level, does not require token-level language modeling in the reasoning loop — and the decomposition that makes this possible also makes the reasoner mode-agnostic and lets scale be spent on thinking rather than on surface statistics (§7). The codec's importance-ordered, prefix-decodable latents make compression a decode-time dial; the thinker's winner-take-all objective with cycle consistency keeps its predictions decodable without collapsing them.

The scale is also small enough that a full ablation round costs an afternoon, which is precisely what made the register case study possible: we could afford to run every hypothesis to ground, trace a behavioral disease to a data-distribution absence, attempt the data patch, and observe that the patch bought partial, topic-gated routing rather than the underlying skill. We release code, logs, and checkpoints so that these results can be reproduced — and audited — on hardware anyone can own.

History of this project

This idea had been on my mind since sometime in 2024, but I didn't yet have the skills to build it. As I grew, and as LLMs grew in capability, I kept returning to it: every few months I would test whether a model could understand the system and was capable enough to implement it.

When I gave it a shot with Claude Opus in early 2026, I caught a glimmer of hope, but it was limited. The codec pair lost stability on longer sequences, maxing out around 100 tokens before complete collapse into gibberish — much like an earlier attempt built with DeepSeek V4 — even though, architecturally, it could handle sequences up to 256 tokens. The efficiency was also questionable, needing roughly one vector per 1.8 tokens for byte-perfect representation. The DeepSeek-era prototype's research log, preserved at `legacy/phase0/RESEARCH.md` in the repository, records these limits quantitatively: near-perfect reconstruction held only below $\sim$60 tokens, and past $\sim$80 tokens word overlap plateaued at 50–62% no matter how many vectors were supplied. It felt like there was something here, but Opus felt limiting on what to train on, how to train, and so forth. So I waited several more months.

Then Claude Fable released in early June. I gave it another shot, and with its insights, broad knowledge, and brutal iterative capability we created the codec pair this project uses. It represented sequences as long as the architecture allowed (we didn't go larger due to hardware limitations), and it did shockingly well: almost always byte-perfect reconstruction at one vector per four tokens, with graceful, grammatically correct degradation at one per eight and beyond — proving it isn't simply passing the raw tokens through.

We then moved on to building the thinker, but that work stopped abruptly when access to Fable was restricted in the US. I tried to continue with Opus, with results much like before — something interesting, nothing concrete. It wasn't until July 1st, when Fable was available again, that a few days of concentrated work on the problem produced the thinker model this paper documents.

Author and AI contribution statement

This project is a human–AI collaboration where "we" throughout this paper means exactly that. The core concept — dialogue conducted in an ordered, prefix-decodable sentence-latent space — its persistence across three generations of attempts, the project's direction, and the hardware are the human author's. Nearly everything downstream of that is AI work, done mainly by instances of Claude Fable under the author's direction: the implementation, the training recipes, the design and adjudication of the ablation rounds, the metrics and their lexicons, the transcript audits and battery scoring, and the drafting of this paper. Where the paper says a "hand audit" or "hand-scored", that means an AI instance reading transcripts — in contrast to the automated lexicon scoring — not a human rater; §8 weighs what that costs the evaluation. The author did not simply take the AI's word for the result: after the AI judged the final checkpoints good, the author verified the headline claims in live conversation with the released models. The intermediate audits, however, are AI-only, and the repository's logs and transcripts exist so that no reader has to take either of us on faith.

Acknowledgments

Thanks to Anthropic for creating Claude Fable, which enabled me to pursue this idea, and to my parents for providing the power, the internet, a roof over my head, and so much more that this project needed.

Special thanks. To my family, for not complaining too much about the fans on my computer; to my brother, for dealing with me being "boring" while working at my computer; and to The Church of Jesus Christ of Latter-day Saints, for help during a mentally difficult time.

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A Reproducibility

All code, configs, eval scripts, and the complete experimental log are at `https://github.com/nochinator/thought-vectors`. Training used fixed seeds per run; every table row maps to a named checkpoint and log file in the repository (`logs/` holds all training logs, eval transcripts, and per-run metrics). The two earlier generations of the project are also preserved: `legacy/phase0/` contains the original prototype and its full research log (the source of the prior-attempt numbers in §3), and `legacy/phase1/` the intermediate rebuild, with its git history in `legacy/phase1-git-history.bundle`.

B Selected transcripts

All transcripts are greedy decoding (temperature 0); full files are in `logs/` in the repository.

FINAL_12H, multi-turn eval, conversation 1

The flagship at its best and worst in four turns: grounded empathy, then a register error on turn 3 (`logs/FINAL_12H.multiturn.txt`).

```
user > i'm feeling really overwhelmed with work lately.
bot  > I can imagine. What's been going on?
user > my boss keeps piling on deadlines.
bot  > That sounds like a lot of pressure. Have you been
       able to talk to anyone else?
user > i haven't slept well all week.
bot  > That's good to hear. Have you been able to talk to
       anyone about how you're feeling?
user > what should i do?
bot  > We'll see. Just be careful and don't give up on your dreams.
```

FINAL2_12H, register probes: hits and lexicon misses

From `logs/FINAL2_12H.register.txt`. The first two are genuine wins (the second is the first correct contextual reversal any checkpoint produced); the last two are register errors the lexicon scored as clean — incident (4) in §6.4.

```
user > my dog passed away last night.
bot  > Oh no, I'm sorry to hear that. What happened?
------
user > my sister's wedding planning has been so exciting!
     / but our grandmother is too ill to attend now.
bot  > oh no. what happened?
------
user > i twisted my ankle and can't play soccer for a month.
bot  > How are you going to celebrate?
------
user > i burned the dinner i spent all day cooking.
bot  > Nice, what did you do?
```

FINAL2_12H, live chat probe: the template, not the skill

A reversal conversation with the same shape as the training splices but novel phrasing (`logs/FINAL2_12H.chatprobe.txt`). The probe suite above is defeated; Appendix C bounds how much of the behavior is actually present.

```
user > i just got back from an amazing beach vacation!
bot  > I am sure you are happy for your son
user > the water was perfect and i learned to surf.
bot  > I am sure you will be okay, you will have a great time!
user > but when i got home i found out my apartment was broken into.
bot  > I bet that was fun!
user > they took my laptop and my grandmother's necklace.
bot  > That's a good idea. What did you do?
```

R7 all-hypotheses dump (disease localization)

After an upbeat turn 1, decoding *all four* WTA hypotheses on a contextual bad-news probe yields four *positive* candidate replies, in both FINAL_12H and R7_EMP1X — no selection rule can fix contextual register, because no correct hypothesis exists to select (diagnostic (iii), §6.2; full dump in the experimental log, 2026-07-02 entry).

C Audit of the audit: the ten-reversal battery

The template-not-skill verdict above originally rested on a four-conversation chat probe. During release preparation, fresh two-turn reversals with novel phrasing came out *correct* on FINAL2_12H, so we widened the audit: ten scripted good-news→bad-news conversations with novel topics, pivot at turns 2–4, and pivot cue varied ("but", "unfortunately", none), plus four sustained-mood controls (`scripts/register_battery.py`; full transcripts in `logs/FINAL_12H.battery.txt`, `logs/FINAL2_12H.battery.txt`, `logs/FINAL2_12H.battery_affinity.txt`, `logs/b3_lm_48m_24h/register_battery.txt`). Pivot-turn replies, hand-scored at temperature 0:

Checkpoint	Selection	Commiserates	Cheerful (disease)	Neutral/confused
FINAL_12H	decodable	0/10	7	3
FINAL2_12H	decodable	3/10	5	2
FINAL2_12H	affinity	4–5/10	2	3–4
token LM 24h (§6.5)	—	1/10	7	2

Table 5: The reversal battery. FINAL_12H never commiserates after an upbeat turn; FINAL2_12H does so on its patch data's dense distress topics (illness, job loss, a lost pet) and still cheers situational or property misfortune (a hailstorm, a rear-ending, the archived burglary). The matched token LM lands at FINAL_12H's level — its one hit is the battery's most distress-dense topic (job loss) — so the splice-trained thinker remains the best register router tested. Sustained-mood controls stay correct for all models.

Representative pivot replies (FINAL2_12H, decodable): "but my knee scans came back and i need surgery" → "That's terrible. [. . .]" (hit); "a hailstorm destroyed all of it last night" → "I love the new one!" (miss); "then on my drive home a truck rear-ended me" → "Oh, I bet that was fun." (miss). The same conversations on FINAL_12H yield no hits at all ("i lost my job this morning" → "i love to be a good person .”). Affinity selection corrects the archived burglary probe ("I'm sorry to hear that. Did you get it back?") but not the rear-ending. The pattern — routing succeeds exactly where EmpatheticDialogues-style distress vocabulary is dense — is what grounds the "partial, topic-gated" characterization in §6.2.

D Ablation tables: the register rounds

Canonical register scores for all round 6–8 arms, rescored under the final (v3) lexicon so rows are comparable; every arm is a 45-minute warm-start from FINAL_12H. The complete per-round tables (rounds 3–8, all metrics) are in `RESEARCH_LOG.md` in the repository.

Round	Arm	reg↓	reg_ctx↓	pos_ok↑
—	FINAL_12H (base)	0.17	0.50	0.17
R6	control (more base data)	0.25	0.83	0.33
R6	cycle_frac 0.5	0.33	0.83	0.67
R6	w_cycle 1.0	0.25	0.83	0.33
R6	8 hypotheses	0.33	0.67	0.33
R6	unfreeze codec	0.17	0.67	0.33
R7	+ED 1×	0.42	0.83	1.00
R7	+ED 2×	0.42	1.00	0.83
R7	+ED 1×, no cycle	0.42	0.83	1.00
R8	+20k reversal splices	0.67	0.83	1.00
R8	+40k reversal splices	0.67	0.50	0.83

Table 6: Register metrics, v3 lexicon, temperature 0. Every fine-tuning arm — including the do-nothing control — worsens contextual register; the ED arms buy positive-affect style (pos_ok) at the cost of routing (reg); only the from-scratch FINAL2_12H run (Table 2) moves reg_ctx below base, and §6.2 qualifies that win.